

[Hospital Logo]

NURSEQUEST

PROGRESS UPDATE & NEW FEATURES

*Overview of Curriculum Organization, Advanced Quiz Question Types, and UX Enhancements
Added Post-Web-Preview*

JUNE 9, 2026

SREC · TECHNICAL PLATFORM DEVELOPMENT

EXECUTIVE SUMMARY

Since the publication of the initial 'NurseQuest Web Preview' document, the platform has undergone significant architectural expansion and user-experience polishing. This Progress Update documents the features, pages, database schema modifications, and question formats implemented since then to transition the project from a basic quiz system into a full-scale, structured clinical training environment.

What is NOT in the original Web Preview document:

- **Interactive Landing Page & Guest Join Flow:** A pre-authorization homepage with theme toggles, detailed Host vs. Player experience grids, and an interactive phone mockup demonstrating friction-free 6-digit room PIN joins.
- **Curriculum-Based Learning Modules:** A core hierarchy grouping quizzes into clinical subjects (e.g., Fundamentals of Patient Care, Pharmacology) rather than simple text categories. Includes a dedicated Module Manager for teachers and Module View with progress indicators for students.
- **Advanced Question Formats (Slider & Matching Pairs):** Moving beyond multiple choice to support clinical vital range inputs (via Sliders) and term relationships (via Matching Pairs) both in single-player and Socket.io multiplayer.
- **Light / Dark Theme Support:** Full stylesheet token integration supporting real-time visual mode switches.
- **Expanded Database Schema:** Creation of the `modules` relational table and extension of the `questions` table with fields for slider constraints and matching pairs JSON arrays.

1. INTERACTIVE LANDING PAGE & GUEST JOIN FLOW

A major UX gap in the original web preview was the lack of an introductory page. The platform now features a high-impact landing page at the root route (`/`) which introduces the application to guests, explains the features, and provides a comparative review of Host vs. Player experiences.

Core Enhancements on this Page:

- **Guest Joining:** Students do not need to register. They can click 'Enter Code' directly on the landing page, opening a modal to submit a 6-digit room PIN and choose a nickname.
- **Nickname Generator:** A casino-style random nickname generator is built into the join flow, combining safe prefixes and clinical terms (e.g., 'MightyPulse38') and checking against a blocked list of words to ensure classroom safety.
- **Theme Toggle:** A navbar controller allows users to switch between Light Mode and a sleek dark layout instantly.
- **Parallax Design:** Interactive background blur blobs track mouse movements, giving the landing page a premium, dynamic feel.

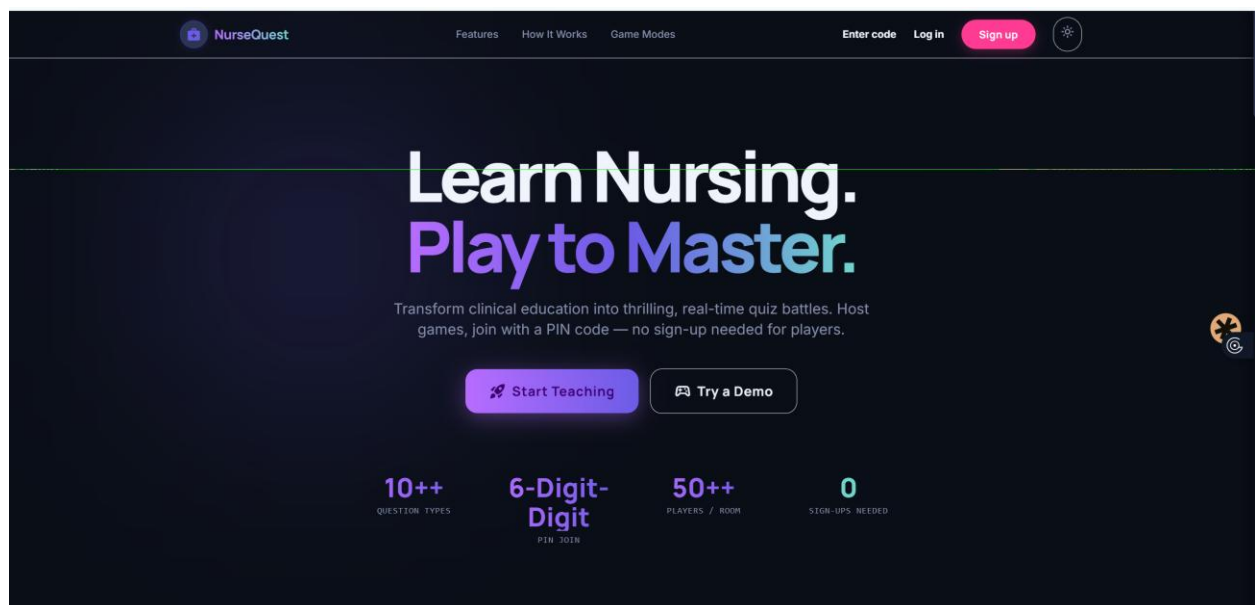


Figure 1: Interactive Landing Page (/) featuring phone join mockup, theme switch, and statistics counters.

2. LEARNING MODULES — TEACHER VIEW

To structure learning pathways, we implemented a relational Learning Modules system. Instead of quizzes being loosely categorized by text strings, they are now child elements of defined Course Modules. Teachers manage this via the new Module Manager (`/modules`).

Module Manager Features:

- **Module Creation & Metadata:** Teachers can define a Module Title and Description, select an appropriate icon (from 12 Material Icons), and choose a unique accent color (from 10 pre-curated color tokens) to color-coordinate the module's styling.
- **Drafting and Publishing:** Toggle switches determine whether a module is a Draft (invisible to students) or Published (live on student dashboards).
- **Drill-Down Quiz Management:** Expanding a module's row displays all quizzes assigned to that module, their draft/published states, question counts, and allows direct quiz editing or previewing.

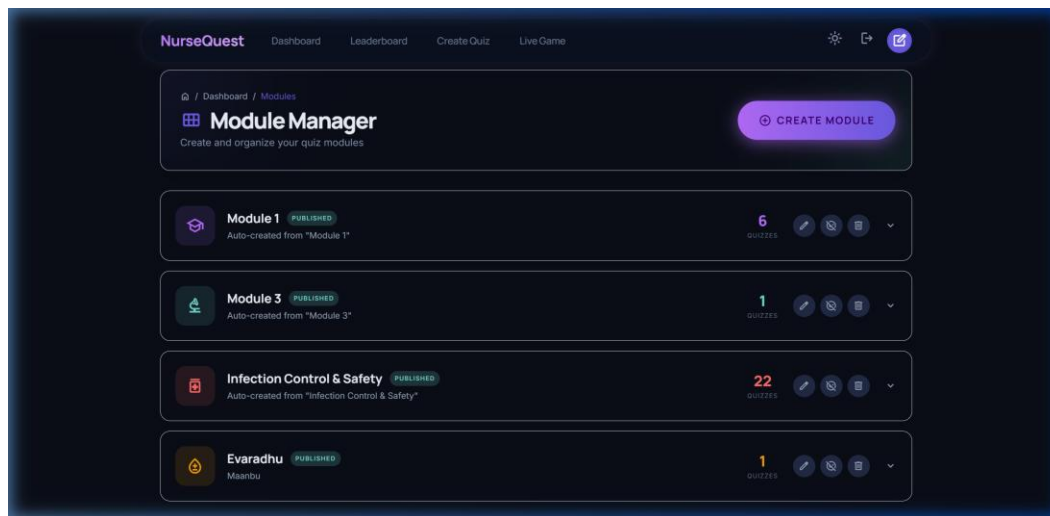


Figure 2: Teacher Module Manager (/modules) displaying the interactive module editor modal and quiz drill-down.

3. LEARNING MODULES — STUDENT VIEW

For students, modules provide a structured, gamified learning map. When a student clicks on a module from their dashboard, they are navigated to the dedicated Module View (`/modules/:id``), styled with the module's specific accent color.

Student Module Cockpit Features:

- **Animated Progress Tracking:** A linear progress bar dynamically calculates and displays the percentage of quizzes completed within the module (e.g., '2/3 Quizzes Completed · 67%').
- **Visual Completion States:** Quizzes that have been attempted show a checkmark status icon and display the student's highest score. Unattempted quizzes show a numeric index and a 'Start' CTA.
- **Color-Themed UI Glows:** The entire UI adaptively updates its buttons, progress fills, and background glows to match the accent color designated by the teacher for that specific module.

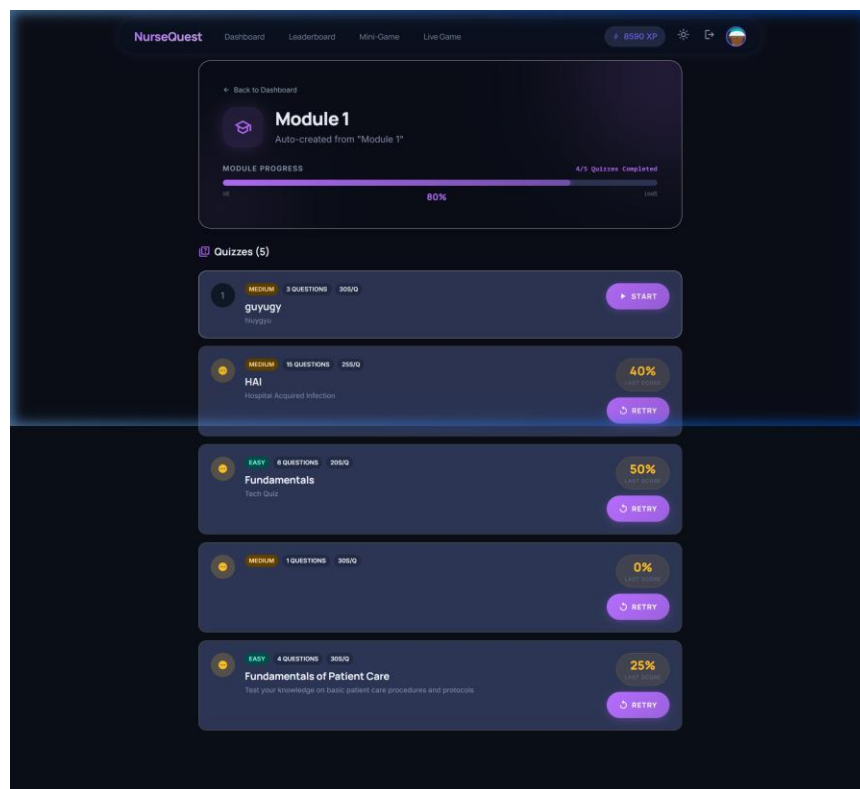


Figure 3: Student Module View (`/modules/:id``) showing the color-coordinated progress bar, completion ticks, and quiz list.

4. CLINICAL QUESTION TYPES

To better simulate clinical scenarios, the Quiz Builder and Player have been expanded with two advanced question types: Sliders and Matching Pairs. These allow nursing students to interact with continuous numerical vital ranges and relational data.

Enhancement A: Slider Questions (Continuous Numeric Input)

Used for questions requiring numeric values (e.g. vital signs, drug dosages). Teachers configure the minimum value, maximum value, step size, unit (e.g., 'mmHg', 'bpm', '°C'), and the correct numeric answer. Students drag a custom-styled slider to submit their answer, which is checked against the database.

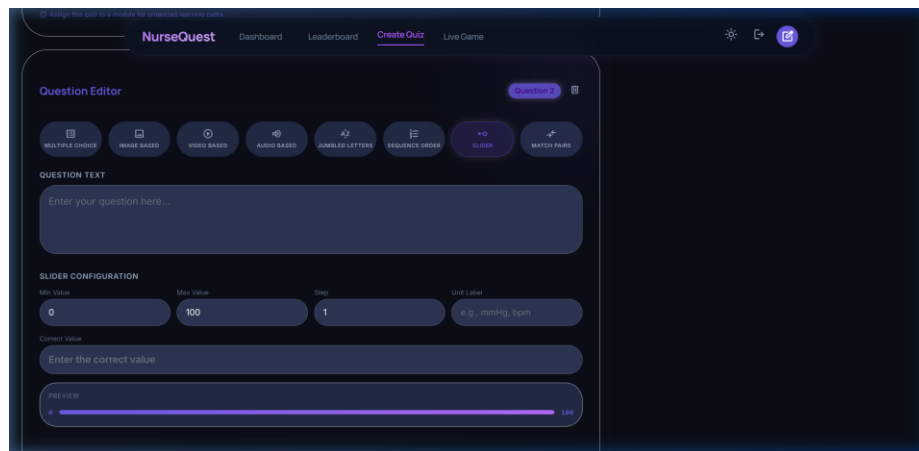
The screenshot shows the 'Question Editor' interface in the 'NurseQuest' application. The 'Create Quiz' tab is active. In the 'Question Editor' section, the 'SLIDER' question type is selected. The 'QUESTION TEXT' field contains the placeholder 'Enter your question here...'. The 'SLIDER CONFIGURATION' section includes fields for 'Min Value' (0), 'Max Value' (100), 'Step' (1), and 'Unit Label' (e.g., mmHg, bpm). Below these is a 'Correct Value' field with the placeholder 'Enter the correct value'. At the bottom, there is a 'PREVIEW' section showing a visual representation of the slider with a red bar and a blue handle.

Figure 4: Quiz Builder — Slider Question configuration interface (setting min, max, step, unit, and answer).

Enhancement B: Matching Pairs (Relational Clinical Mapping)

Used to map symptoms to clinical diagnoses, or drugs to side effects. Teachers add list pairs. Students click left-side blocks and right-side blocks to form pairs, rendered with visual connection states. The scoring engine verifies all pairs on submit. Fully supported in both self-paced and live multiplayer.

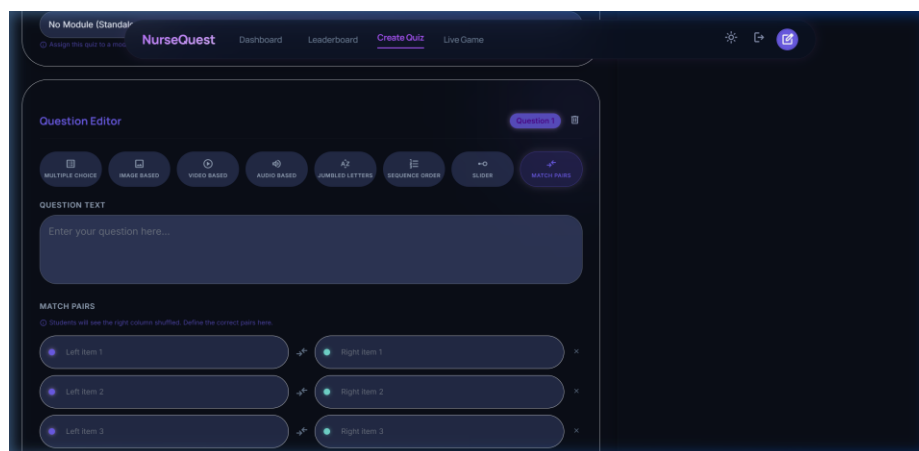
The screenshot shows the 'Question Editor' interface in the 'NurseQuest' application. The 'Create Quiz' tab is active. In the 'Question Editor' section, the 'MATCH PAIRS' question type is selected. The 'QUESTION TEXT' field contains the placeholder 'Enter your question here...'. The 'MATCH PAIRS' section includes a checkbox for 'Students will see the right column shuffled' and a table for defining pairs. The table has two columns: 'Left Item' and 'Right Item'. Three rows are shown, each with a 'Left Item' and a 'Right Item' field, and a 'Match' button between them.

Figure 5: Quiz Builder — Matching Pairs Question creation section.

5. DATABASE SCHEMA UPDATES

To support these features, the SQLite schema was expanded from the original 9 tables to 10 tables, and new columns were appended to existing tables. Below is the updated relational structure.

Table Name	Structure & Foreign Keys	Added Columns / Purpose
Table Name	Structure & Primary Key	Added Columns / Purpose
modules	id (TEXT PK)	title (TEXT), description (TEXT), icon (TEXT), color (TEXT), order_index (INT), created_by (TEXT FK), is_published (INT). Groups quizzes into clinical categories.
quizzes	id (TEXT PK)	Added module_id (TEXT FK to modules). Links quizzes dynamically to modules rather than text categories.
questions	id (TEXT PK)	Added type check: allows 'slider' and 'matching'. Added: slider_min (REAL), slider_max (REAL), slider_step (REAL), slider_unit (TEXT), matching_pairs (TEXT - JSON).

Migration and Seed Integrations:

- **db/migrate_modules.js:** A script to auto-generate the `modules` table, alter `quizzes` to add the foreign key, and cluster old loose quizzes into default modules.
- **db/migrate_new_types.js:** Re-creates the `questions` table schema with the slider-min/max/step/unit/matching-pairs columns and a CHECK constraint allowing 'slider' and 'matching' types.
- **db/seed.js:** Seeds three modules (Fundamentals, Anatomy & Physiology, Pharmacology) and populates multi-format questions (MCQ, Image, Video, Audio, Jumbles) with achievements and mock leaderboard data.

6. Bug Fixes(Patches) and improvements:

- Now the Multimedia(audio, video, image) questions will work properly and Teachers can choose file from their local directory
- Live quiz session's timers are fixed
- Theme is introduced
- Login attempts in same computer does not direct you to the same page in another tab. Each time you have to login.
- '+' button in the top right corner is changed into a create quiz button